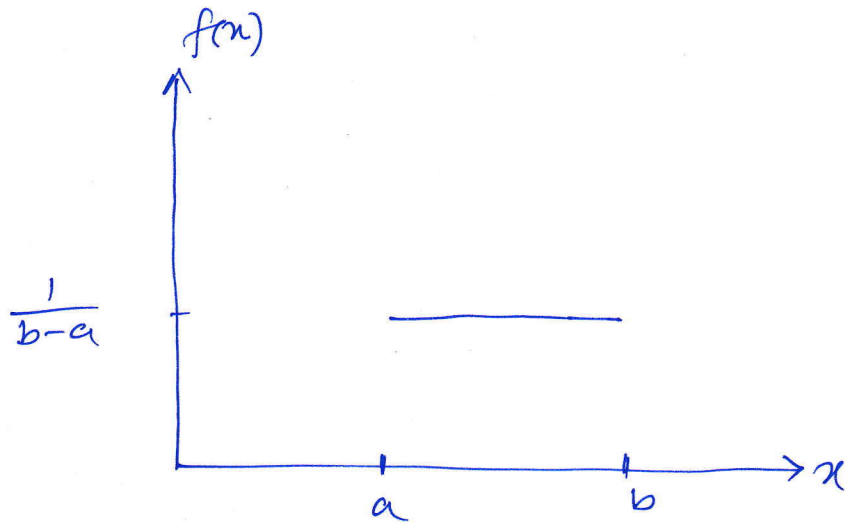


Uniform Distribution : has constant probability within a certain domain



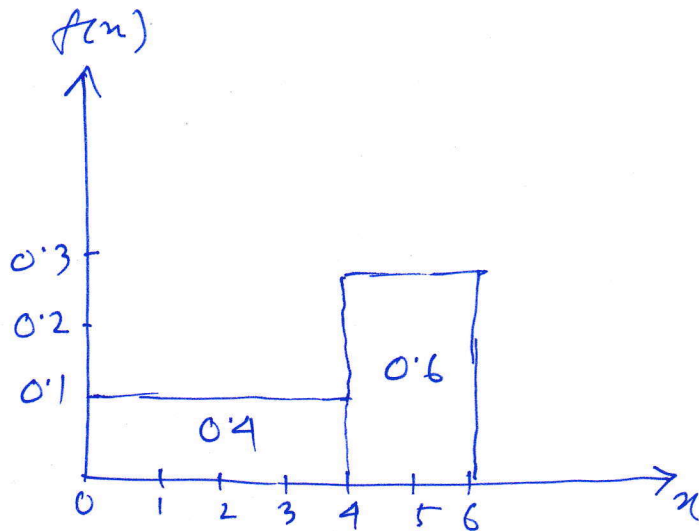
$$f(x) = P(a \leq x \leq b) = 1 = \text{height} \cdot \text{width}$$
$$1 = \text{height} \cdot (b-a)$$
$$\text{or } \frac{1}{b-a} = \text{height}$$

Range : $[a, b]$

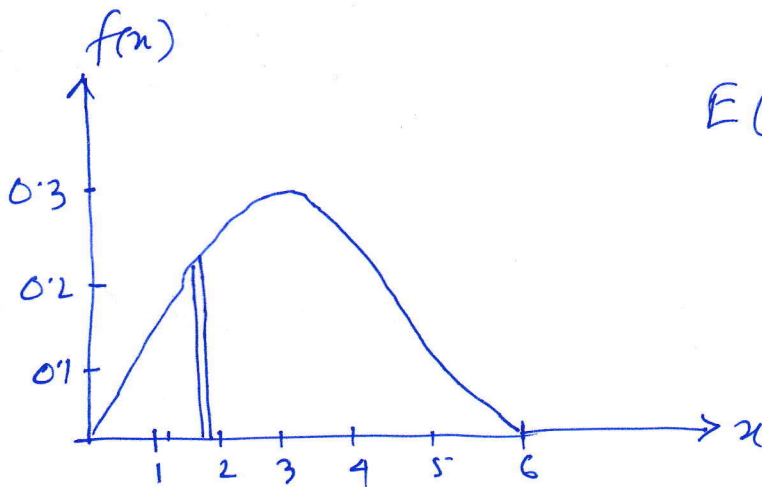
$$\text{Mean } (\mu) : \frac{b+a}{2}$$

$$\text{Variance} : \frac{(b-a)^2}{12}$$

Expected Value: Calculated from probability distribution.



$$E(X) = 0.4 \times 2 + 0.6 \times 5 \\ = 3.8$$



$$E(X) = \int_0^6 x \underline{f(x)} \underline{dx}$$

Mean (μ): Calculated from occurrence distribution.

Variance:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \\ \approx E(X^2) - \mu^2$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \int_{-\infty}^a x^2 f(x) dx + \int_a^b x^2 f(x) dx + \int_b^{\infty} x^2 f(x) dx$$

$$= 0 + \int_a^b x^2 \cdot \frac{1}{b-a} dx + 0$$

$$= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b$$

$$= \frac{1}{3(b-a)} (b^3 - a^3)$$

$$= \frac{1}{3(b-a)} (b-a) (b^2 + ba + a^2)$$

$$= \frac{b^2 + ba + a^2}{3}$$

$$M^2 = \left(\frac{b+a}{2} \right)^2$$

$$= \frac{b^2 + 2ba + a^2}{4}$$

$$\text{Var}(X) \approx E(X^2) - M^2$$

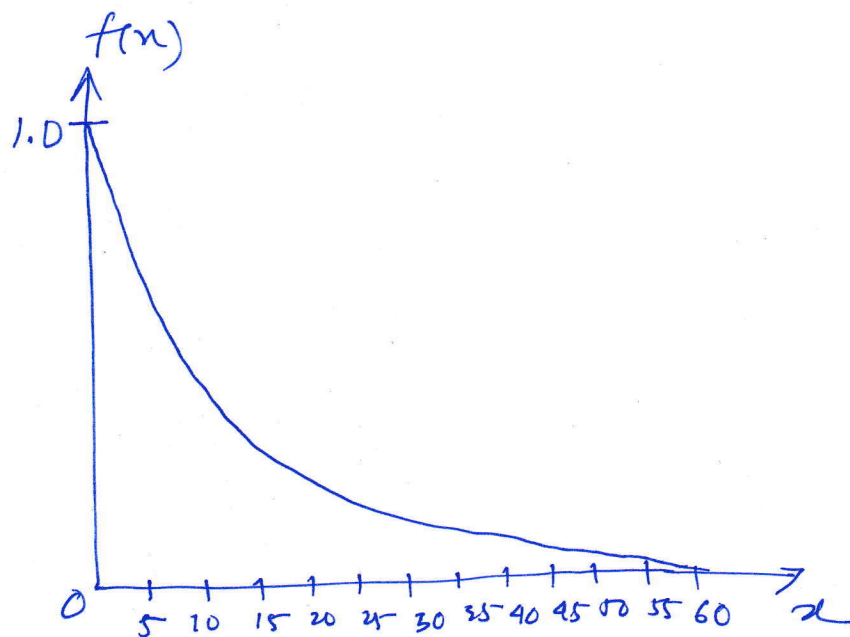
$$= \frac{b^2 + ba + a^2}{3} - \frac{b^2 + 2ba + a^2}{4}$$

$$= \frac{4b^2 + 4ba + 4a^2 - 3b^2 - 6ba - 3a^2}{12}$$

$$= \frac{b^2 - 2ba + a^2}{12}$$

$$= \frac{(b-a)^2}{12}$$

Exponential Distribution: If a website is visited by 10 users per hour, then minutes until the next user will visit...



$$f(x) = \begin{cases} \frac{1}{\beta} e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

[β is maximum likelihood of the function]

Expected

value:

$$E(x) = \int_0^{\infty} x f(x) dx$$

$$= \int_0^{\infty} x \frac{1}{\beta} e^{-x/\beta} dx$$

$$= \frac{1}{\beta} \int_0^{\infty} x e^{-x/\beta} dx$$

$$= \frac{1}{\beta} \left\{ \left[x \cdot \frac{e^{-x/\beta}}{-\frac{1}{\beta}} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-x/\beta}}{-\frac{1}{\beta}} \cdot \frac{dx}{dx} \cdot dx \right\}$$

$$= \frac{1}{\beta} \left\{ \left[\frac{\infty \cdot e^{-\infty/\beta} - 0 \cdot e^{0/\beta}}{-\frac{1}{\beta}} \right] + \beta \int_0^{\infty} e^{-x/\beta} dx \right\}$$

$$= \frac{1}{\beta} \left\{ 0 + \beta \left[\frac{e^{-x/\beta}}{-\frac{1}{\beta}} \right]_0^{\infty} \right\}$$

$$= \frac{1}{\beta} \cdot \beta \left[\frac{e^{-\infty/\beta} - e^{-0/\beta}}{-\frac{1}{\beta}} \right]$$

$$= -\beta [0 - e^0]$$

$$= -\beta \cdot -1$$

$$= \beta$$

$$\text{Variance: } \beta^2$$